

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
Kennedy Space Center (KSC)
JUSTIFICATION FOR OTHER THAN FULL AND OPEN COMPETITION
(JOFOC)
For Event Sequence Diagrams (ESD) for the Nuclear Rosalind Franklin Mission
(RFM) Launch Vehicle Risk Assessment

1. **Federal Acquisition Regulation (FAR) 6.104-1(a)(1) – Identification of the agency and the contracting activity, and specific identification of the document as a “Justification for other than full and open competition.”**

This document is a justification for other than full and open competition prepared by National Aeronautics and Space Administration (NASA) KSC.

The procuring agency is NASA, and the contracting activity is KSC.

2. **FAR 6.104-1(a)(2) – The nature and/or description of the action being approved:**

This document outlines the rationale for awarding a sole-source purchase order to Advanced Systems Concepts Associates, Inc. (ASCA) for Event Sequence Diagrams (ESD) populated with nuclear-enabled mission specific launch vehicle accident failure scenario and outcome services which characterize risks for all phases of flight. The ESDs deliver essential information on launch vehicle failure mechanisms, their causes, and resulting consequences. This analysis supports the characterization of launch failure probabilities, quantification of accident environments, assessment of inertia effects on nuclear payloads, and determination of nuclear material release risks. Additionally, it enables the identification of risk tiering levels in accordance with National Security Presidential Memorandum (NSPM)-20.

3. **FAR 6.104-1(a)(3) – A description of the supplies or services required, to meet the Agency’s needs (including the estimated value):**

NASA's Launch Services Program (LSP) requires the development of ESDs to assess risk levels for launch vehicles carrying nuclear payloads. For these missions, ASCA provides specialized expertise in NASA launch systems by utilizing confidential, proprietary analytical framework and dataset, in which they own, that incorporate ESDs populated with nuclear launch-vehicle accident failure scenarios and outcomes. Under this purchase order, ASCA shall provide all required services and associated support in fulfillment of the RFM mission, scheduled for launch aboard the SpaceX Falcon Heavy launch vehicle. The Contractor shall perform these efforts over an estimated 13-month period of performance commencing on the date of award for an estimated value of \$563,000.00. Obtaining accurate data to characterize the launch vehicle and spacecraft systems is essential for multiple reasons, including enabling more robust safety and risk analyses related to site selection, range operations, public safety, nuclear payload safety, launch approval processes, and the determination of programmatic requirements for launch license applicants.

4. FAR 6.104-1(a)(4) – An identification of the statutory authority permitting other than full and open competition:

The statutory authority permitting other than full and open competition is 10 U.S.C. 3204(a)(1), as implemented by FAR 6.103-1 Only one responsible source and no other supplies or services will satisfy agency requirements.

5. FAR 6.104-1(a)(5) – A demonstration that the proposed contractor’s unique qualifications or the nature of the acquisition requires use of the authority cited:

The rationale supporting the use of 10 U.S.C. 3204(a)(1) is based on ASCA’s specialized expertise and proven capabilities, demonstrated through their execution of ESDs for numerous NASA missions, including Cassini, Mars Pathfinder, Mars Exploration Rover A and B, Pluto New Horizons, MSL, Mars2020, and DragonFly.

ASCA provides a comprehensive data product encompassing methodology, proprietary tooling, and extensive historical flight and anomaly records specific to NASA. By integrating branch point probabilities into the ESDs, ASCA enables the development of an independent approach to assessing the likelihood of launch vehicle failure relative to mission elapsed time and overall reliability. Their proprietary process provides very precise level of precision and mission-specific ESDs which are supported by a global database of launch vehicle histories that is unmatched by any other entity. These distinct competencies are critical to the NSPM-20 nuclear launch approval process.

6. FAR 6.104-1(a)(6) – A description of the efforts made to ensure that offers are solicited from as many potential sources as practicable, including whether a notice was or will be publicized as required by part 5 and, if not, which exception under part 5 applies:

A notice to the Government Point of Entry (GPE) website (Sam.gov) was published on February 12th, 2026 in accordance with FAR part 5. This posting informed potential sources of NASA’s intent to award this sole-source Purchase Order to ASCA, Inc. The results of this synopsis are summarized in Section 8 below.

7. FAR 6.104-1(a)(7) – A determination by the contracting officer that the anticipated cost to the Government will be fair and reasonable:

The Contracting Officer’s (CO) signature on this document indicates that the CO has determined that the anticipated cost to the government will be fair and reasonable. The contractor must be required to submit a proposal to be evaluated and negotiated by the Government. Prior to execution of the contractual instrument, a proposal analysis will be performed. The proposal analysis will ensure that the final agreed-to price for the contract action is fair and reasonable.

8. FAR 6.104-1(a)(8) – Description of the market research conducted, and the results, or a statement of the reason market research was not conducted (e.g., urgent, and compelling requirement):

A notice of the requirement was posted to the GPE website on February 12, 2026 in accordance with FAR Part 5 to determine whether capable sources existed. Two companies submitted capability statements, which were subsequently reviewed by technical. Following this assessment, neither respondent demonstrated the capacity to satisfy NASA's nuclear launch approval standards or to provide the required ESD methodology, proprietary data, or schedule needed for RFM. Furthermore, attempting to re-develop and replicate the specialized expertise and proprietary analytical framework provided by ASCA would extend the schedule significantly, requiring multiple review cycles to produce the desired outcome, and would ultimately result in unnecessary, duplicative costs to the Government. Accordingly, market research verified that no other responsible sources will be capable to fulfill these requirements.

9. FAR 6.104-1(a)(9) – Any other facts supporting the use of other than full and open competition:

None.

10. FAR 6.104-1(a)(10) – A listing of the sources, if any, that expressed an interest in writing in the acquisition:

As mentioned in section 8 above, a notice of NASA's intent to award this sole-source action was posted to the GPE that closed on February 27, 2026. Two companies submitted written capability statements in response: DreamCatchers Consulting, LLC. and Teixido-Boreal Research Group. Both responses were provided to the technical for review. Based on the technical assessment, neither company demonstrated the ability to meet NASA's nuclear launch vehicle accident failure scenario requirements for RFM.

11. FAR 6.104-1(a)(11) – A statement of actions, if any, the agency may take to remove or overcome any barriers to competition before any subsequent acquisition for the supplies or services required:

The Agency will continue to examine the market in the future for alternative solutions or new sources before executing any subsequent acquisitions for the same requirements.

**Launch Services Program
Rosalind Franklin Mission
Launch Vehicle Probabilistic Reliability Risk Analysis Services**

**JUSTIFICATION FOR OTHER THAN FULL AND OPEN COMPETITION
SIGNATURE PAGE**

Technical Representative: I certify that the facts presented in this justification are accurate and complete.

David Piryk Digitally signed by David Piryk
Date: 2026.03.19 10:37:40 -04'00'

David Piryk
Technical Representative

Contracting Officer: I hereby certify that the above justification is complete and accurate to the best of my knowledge and belief.

Allyson Grooms Digitally signed by Allyson Grooms
Date: 2026.03.19 11:32:00 -04'00'

Allyson Grooms
Contracting Officer